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Graded Assignment • 30 min

Due Mar 20, 11:59 PM CET

1. Which of the following is false about the two keys used in public key encryption? 1 point

- ○ One key is public and the other is private
- ○ The public key alone is used for encryption and the private key alone is used for decryption
- ○ The public key is openly revealed while the private key must be kept secret
- ◉ If you have the public key it is easy to compute the private key

2. When you are using secure http and sending data between your computer and your bank's computer, where is the data encrypted and decrypted? 1 point

- ◉ Encrypted in your computer and decrypted in the bank's computer
- ○ Encrypted and decrypted each time your message passes through a router
- ○ Encrypted in your network card and decrypted by the bank's Internet Service Provider (ISP)
- ○ Encrypted in your bank's computer before they store the data in a database

3. In regards to security, what do we assume about the Internet? 1 point

- ◉ That neither the routers nor the links between the routers are secure
- ○ That no one can view Internet traffic without a subpoena
- ○ That the entire Internet is completely secure other than WiFi connections
- ○ No one should ever send sensitive information like Credit Card numbers over the Internet

4. This week we've updated our model of how we communicate information via the internet to add in a fifth mini-layer to the structure in order to protect the confidentiality of transmissions. What is the new list of layers and in what order do we list them? 1 point

- ◉ * Application
  * Secure Sockets
  * TCP
  * IP
  * Link
- ○ * Application
  * TCP
  * IP
  * Packet Encryption
  * Link
- ○ * Application
  * TCP
  * IP
  * Link
  * Dark Fiber
- ○ * ZIP+Secure
  * Application
  * TCP
  * IP
  * Link

5. What is packet sniffing? 1 point

- ◉ Computers watching packets being transmitted across the network in hopes of finding important or valuable data
- ○ When we use detection equipment to see if link speeds are being exceeded
- ○ When a router looks at the first part of packet to to the routing lookup before the entire packet is received
- ○ When a receiving system anticipates a packet before it is received and pre-sends the acknowledgement to save time

6. Which of the following is FALSE about using secure sockets (i.e. https) to send sensitive information like a credit card across the Internet? 1 point

- ○ It requires complex math to decrypt your data
- ◉ It is impossible to decrypt your data
- ○ It requires large amounts of computing power to decrypt your data
- ○ When using an unsecured WiFi, your data is still protected because of https

7. Which of the following is NOT a major threat to your data when using secure sockets? 1 point

- ○ There may be a virus on your computer monitoring your keystrokes
- ○ You can be unknowingly redirected to a server different from the one you would like
- ◉ Someone may see your public key

8. Which of the following is not an equivalent name to 'digital certificate'? 1 point

- ○ Public key certificate
- ◉ Private key certificate
- ○ Digital certificate
- ○ Identity certificate

9. What is a digital certificate? 1 point

- ◉ An electronic document used to give a public key an identity
- ○ An electronic document used to give a private key an identity
- ○ An electronic document used to give both public and private keys an identity
- ○ A real-life document used to certify the security of your computer

10. What is a certificate authority? 1 point

- ○ An entity that tries to ensure secure transactions by attempting to crack encrypted messages online
- ○ An entity that certifies the mathematical relationship between the public and private keys of a system
- ○ An entity that certifies that you have no virus on your computer
- ◉ An entity that certifies the ownership of a public key by the named subject of the certificate

11. Which of the following is NOT an indicator of the effectiveness of Verisign as a certificate authority? 1 point

- ○ Verisign goes to great lengths to store their private keys securely
- ○ Verisign has a high success rate for maintaining secure transactions
- ○ Verisign is used by major manufacturers like Apple and Microsoft
- ◉ Verisign publishes its private keys on a little-known web site only available to key owners

12. How does your computer typically know the public key of a certificate authority during secure communications? 1 point

- ◉ The public key is stored in the operating system by the manufacturer before you buy or install it
- ○ The public key is emailed to the computer the first time you switch it on
- ○ The computer doesn't use a public key
- ○ The public key has to be downloaded when you are setting up the computer's web browser